Stat380Assignment2

James DiPasquale, Tyler Gully, Ahmad Raja

Introduction/Background Go to Code

Insurance pricing depends heavily on understanding which factors actually drive cost. Nittany Valley Insurance Company provided us with policy data and asked us to build a predictive model to predict total insurance charges using linear regression techniques. The dataset includes demographic and behavioral variables such as age, sex, BMI, smoking status, number of dependents, region, claim month, and prior claim activity, along with the total charges billed to each policy.

Our objective is to use this information to build and evaluate predictive models for insurance charges. To do this, we begin with exploratory data analysis to better understand patterns, distributions, and relationships within the data. From there, we develop multiple candidate linear regression models, interpret the estimated coefficients, and assess model performance using appropriate validation methods.

The ultimate goal of this analysis is to recommend the best possible model that is both statistically sound and practically useful for the company's pricing and risk management decisions.

Data Exploration Go to Code

During Exploratory Data Analysis, we used a for loop to graph each predictor variable against charges, the response variable we are trying to predict. After reviewing the initial plots, we determined that categorical variables would be better visualized using boxplots, while continuous variables were more appropriately displayed as scatterplots. We therefore separated the categorical and continuous variables, plotting the categorical variables as boxplots and the continuous variables as scatterplots with a fitted linear regression line to better assess potential relationships. We considered excluding month, sex, and region, as their plots seemed to demonstrate little to no correlation with charges, but decided to include them at the start just in case they are relevant.

Modeling Methods/Strategies Go to Code

Based on the exploratory data analysis, We decided to build a multiple linear regression model starting with a full model using backwards AIC. The reasoning behind this was that several variables were visually identifiable as correlating with charges, but not all of them were very strong. After this succeeded marginally better than the full model, we attempted refinement using a variety of techniques. Backwards BIC, LASSO regressions, and Forward BIC and AIC models all failed to perform better than backwards AIC. We used RMSE to determine model prediction capability.

Results Discussion

A. Initial Results

The initial regression results showed that the model worked pretty well overall and explained a large amount of the variation in insurance charges. With an R^2 of about 0.7211, the model explained about 72% of the variability in charges, which means the predictors did a good job capturing what affects cost. The RSE is only about 10% of the total target variable range, which indicates decent precision. Smoking had by far the biggest impact on charges, while age, BMI, number of children, and prior claims also all had positive effects.

B. Interpretation & Model Evaluation

Looking at the model, all of the predictors except region and month had a positive effect on insurance charges. That means as age, BMI, number of children, and prior claims go up, the predicted charges also go up, assuming the other variables stay the same. Smoking had by far the biggest effect in the model, which shows that it is the strongest factor related to higher insurance costs. The other variables still mattered, but not nearly as much as smoking.

Overall, the model performed pretty well and explained a good amount of the variation in charges. With a relatively low RSE and an adjusted R^2 of 0.7177, the model explained about 72% of the variability in insurance charges, which suggests it did a solid job. The overall model was also highly statistically significant, so the predictors were useful when taken together.

C. Refinements Go to Code

To try to refine the model further, the more selective BIC model was chosen to compare. The notion behind this logic being that BIC is more selective, punishing models with higher predictor counts. This was intended to offer a less complex model to compare to our initial AIC model for evaluating room for improvement. Then, using the test data from the train/test split, we compare each model's ability to predict future data based on their RMSEs.

The initial refinement made a more elegant but less accurate model. While it still explains about the same amount of variation, the backwards BIC model's accuracy on future data is significantly worse than the base full model or the backwards AIC. Thus, we use LASSO to attempt to find a better model. After this failed to perform better, as a last check we performed forward AIC and BIC and compared them with the backward AIC.

No models we found performed better than the initial backwards AIC model.

Final recommendations

Model Summary and Key Findings.

$$\hat{Charges} = 253.70x_{i,age} + 351.27x_{i,bmi} + 456.19x_{i,children} + 22077.38x_{i,smoker} - 837.06x_{i,NWregion} + 0x_{i,NEregion} - 2213.17x_{i,SEregion} - 1582.54x_{i,SWregion} - 114.25x_{i,month} + 483.79x_{i,priorclaims} - 11415.59$$

This model predicts insurance charges based on age, bmi, number of children, smoking status, region, and prior claim count within a residual standard error of about $6,400 and R^2 of .728. This implies that the model explains approximately 73% of the variance in the charges data within an error of about $6,400. According to the linear equation, on average, charges go up by about $262.61 for every year of age, $581.80 for every child, $327.97 for every prior insurance claim, and $23,264.20 for smokers. On average, compared to the Northeast, charges are $1,110.13 less in the Northwest, $2667.75 less in the Southeast, and $2206.32 less in the Southwest. Each month later in the year, charges are on average $109.83 less.

RMSE with Additional Data Go to Code

$$RMSE = 112.6422$$

Given the scale of the target variable and previously derived RMSE values, this is an exceptionally good RMSE, though it is not entirely clear why the model performs better on the validation set than it did on the testing subset created with the train/test split. Possibly the subset was too small. Overall, the candidate model demonstrates strong predictive ability.

Code Appendix

Imports

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
Loading required package: Matrix


Attaching package: 'Matrix'


The following objects are masked from 'package:tidyr':

    expand, pack, unpack


Loaded glmnet 4.1-10

  X caseID age    sex    bmi children smoker    region   charges month
1 1    668  40 female 32.775        2    yes northwest 40003.332     4
2 2    135  20 female 28.785        0     no northeast  2457.211     2
3 3   1295  58   male 25.175        0     no northeast 11931.125     7
4 4    450  35   male 38.600        1     no southwest  4762.329     6
5 5    141  34   male 22.420        2     no northeast 27375.905     9
6 6     12  62 female 26.290        0    yes southeast 27808.725     5
  priorclaims
1           0
```

```
2           0
3           2
4           1
5           1
6           1
```

Data Exploration

```r
cols_num <- which(sapply(insurance_data, is.numeric))

xcont <- list("age", "bmi")

cols_num <- which(sapply(insurance_data, function (x) !(is.numeric(x))))

xcat <- list("sex", "children", "smoker", "region", "month", "priorclaims")

# Excluded Identifier Variables: X, caseid

print(xcont)
```

```
[[1]]
[1] "age"

[[2]]
[1] "bmi"
```

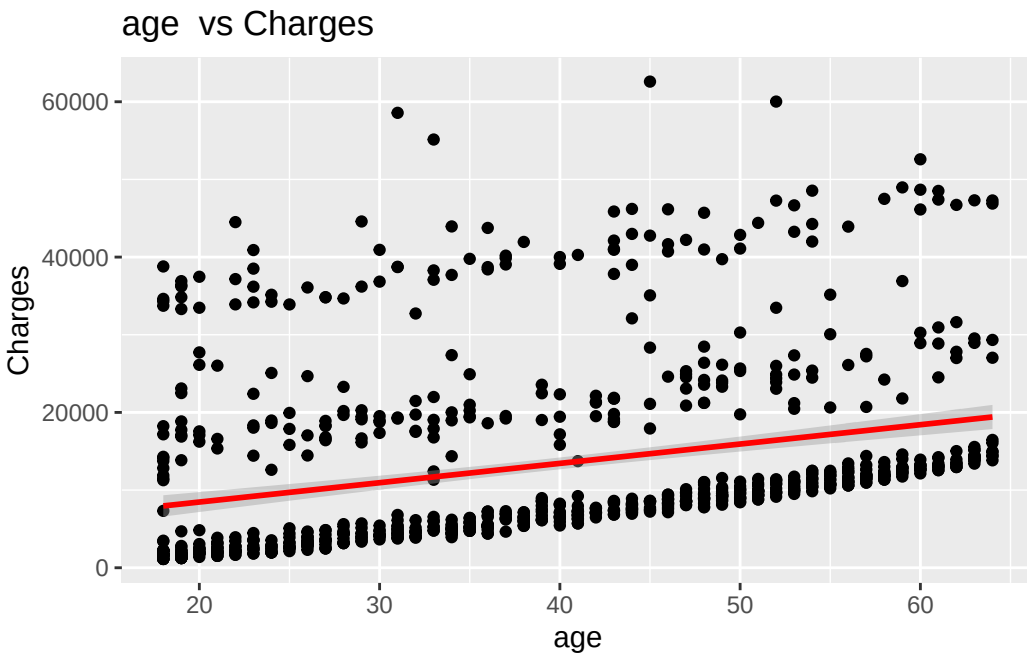
```r
print(xcat)
```

```
[[1]]
[1] "sex"

[[2]]
[1] "children"

[[3]]
[1] "smoker"

[[4]]
[1] "region"
```

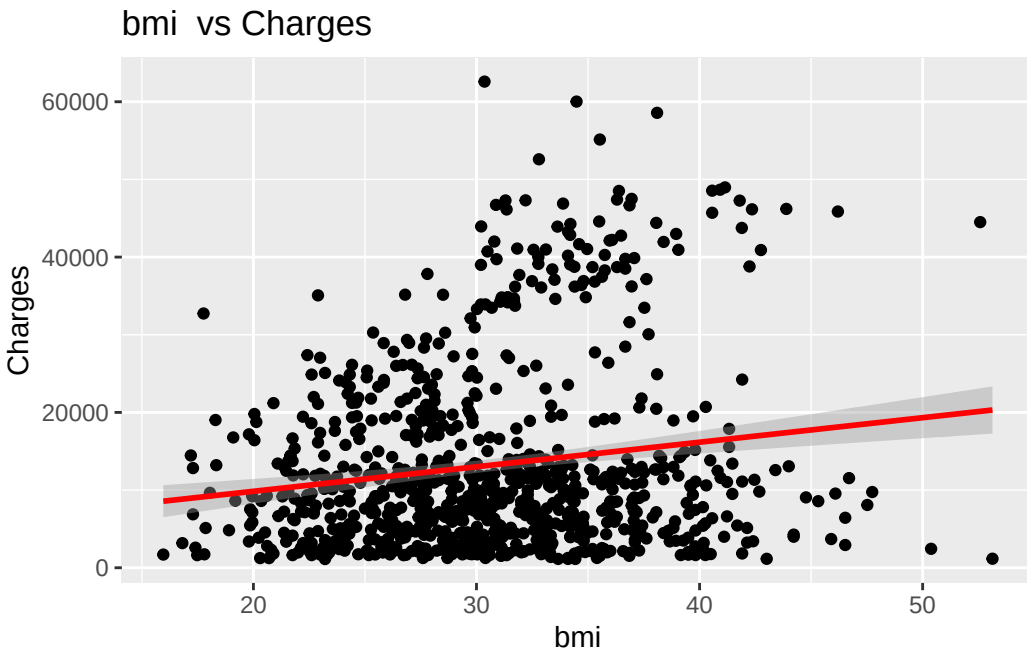
```
[[5]]
[1] "month"

[[6]]
[1] "priorclaims"
```

```r
for (col_name in xcont) {
  plot <- ggplot(insurance_data, aes(insurance_data[[col_name]], charges)) +
  geom_point() +
    geom_smooth(data = insurance_data, method = "lm", color = "red") +
    labs(
    title = paste(col_name, " vs Charges"),
    x = paste(col_name),
    y = "Charges"
  )
  print(plot)
}
```

```
Warning: Use of `insurance_data[[col_name]]` is discouraged.
i Use `.data[[col_name]]` instead.
Use of `insurance_data[[col_name]]` is discouraged.
i Use `.data[[col_name]]` instead.

`geom_smooth()` using formula = 'y ~ x'
```

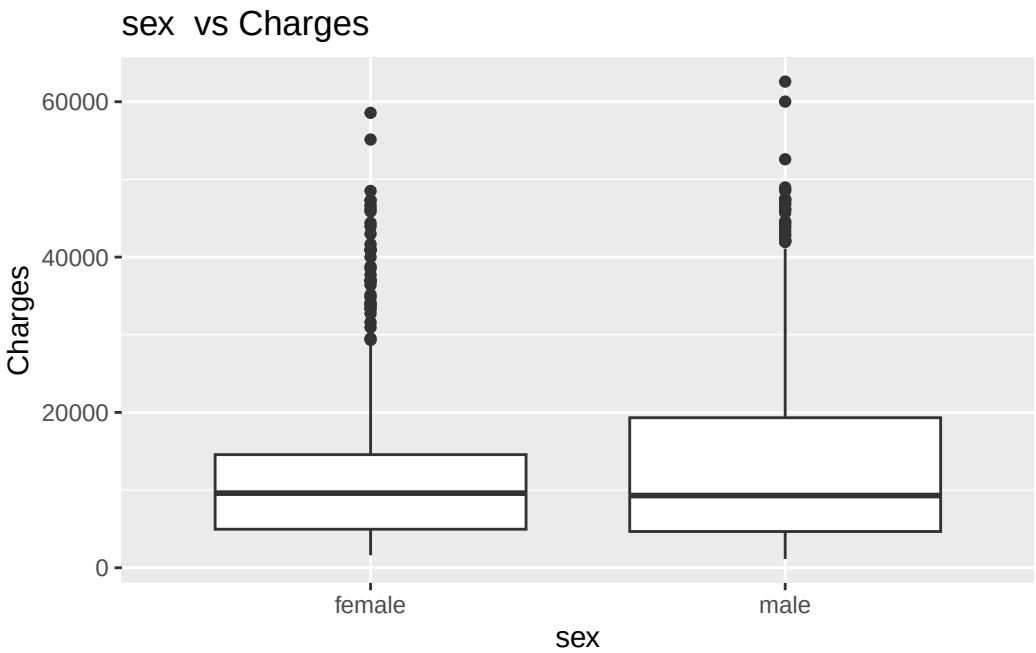



```
Warning: Use of `insurance_data[[col_name]]` is discouraged.
i Use `.data[[col_name]]` instead.
Use of `insurance_data[[col_name]]` is discouraged.
i Use `.data[[col_name]]` instead.

`geom_smooth()` using formula = 'y ~ x'
```

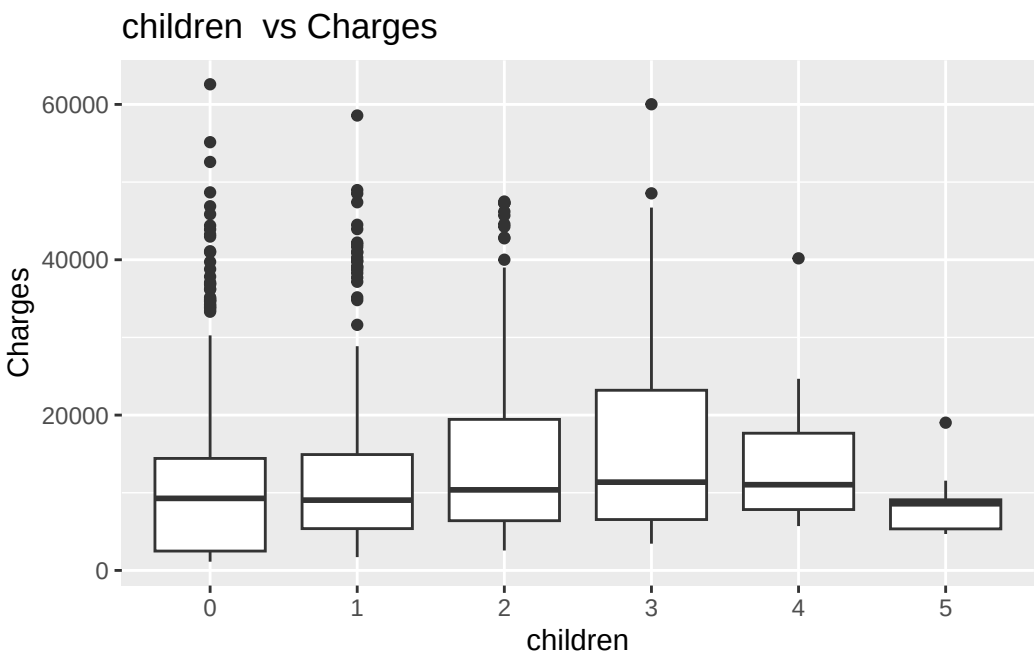



```
for (col_name in xcat) {
  plot <- ggplot(insurance_data, aes(as.factor(insurance_data[[col_name]]), charges)) +
  geom_boxplot() +
    labs(
    title = paste(col_name, " vs Charges"),
    x = paste(col_name),
    y = "Charges"
  )
  print(plot)
}
```

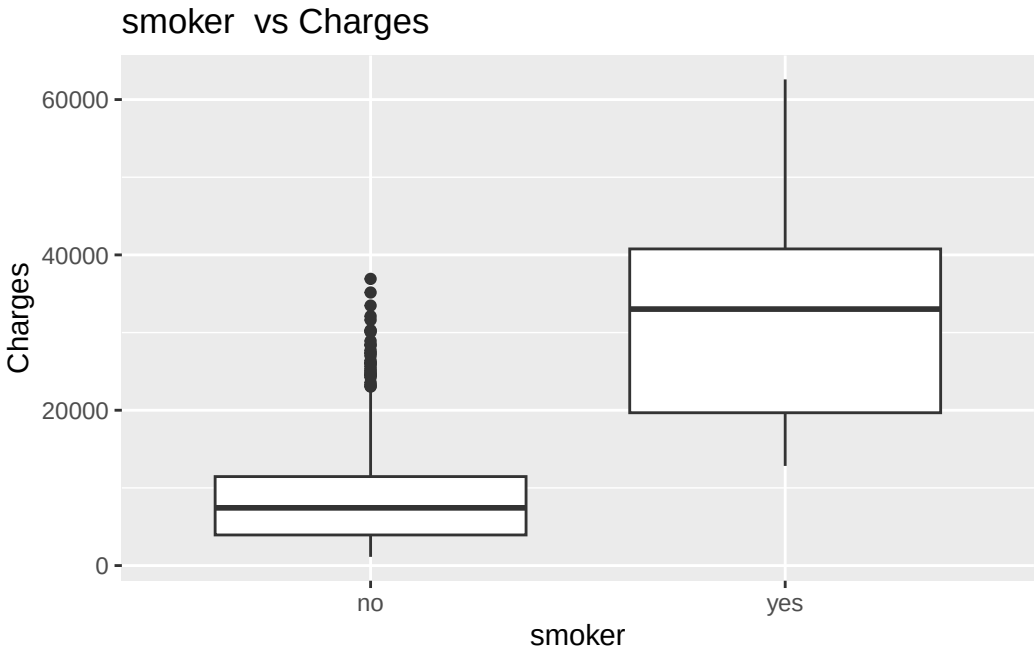
```
Warning: Use of `insurance_data[[col_name]]` is discouraged.
i Use `.data[[col_name]]` instead.
```



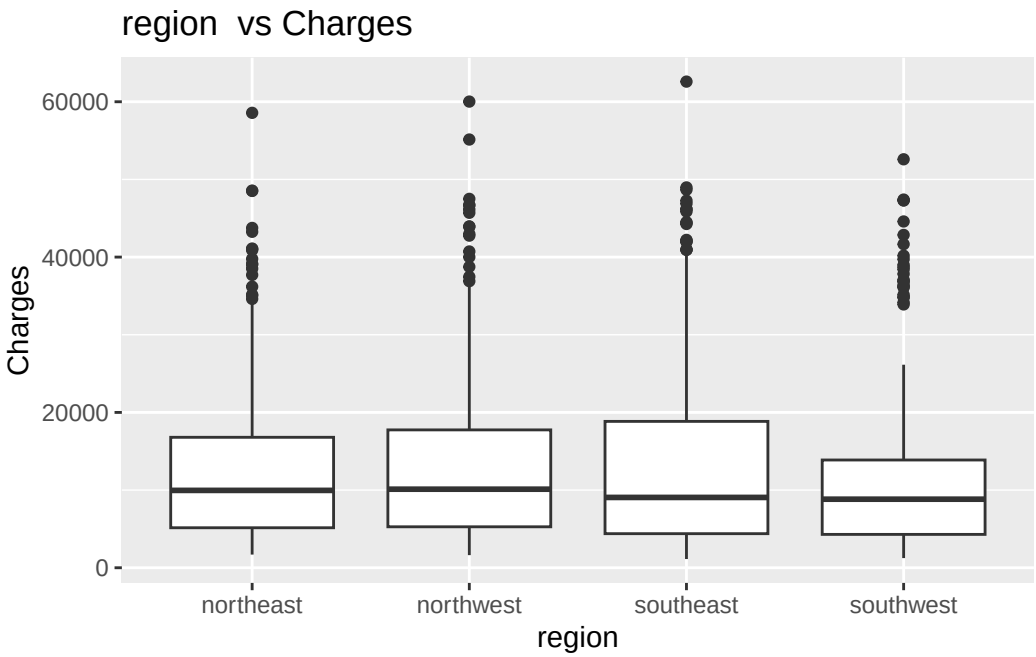

```
Warning: Use of `insurance_data[[col_name]]` is discouraged.
i Use `.data[[col_name]]` instead.
```



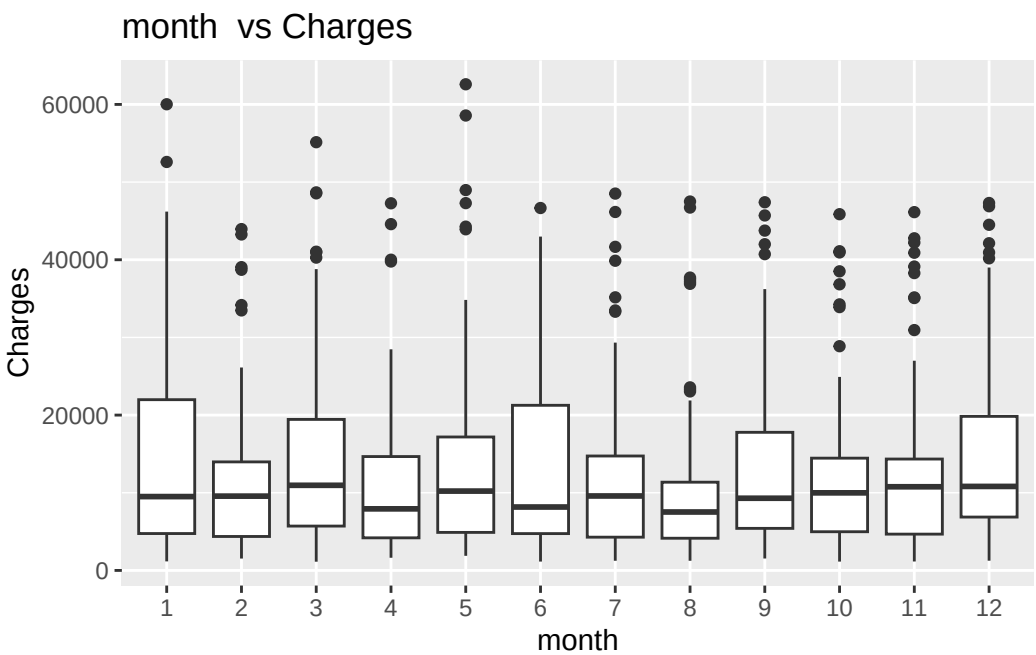
```
Warning: Use of `insurance_data[[col_name]]` is discouraged.
i Use `.data[[col_name]]` instead.
```



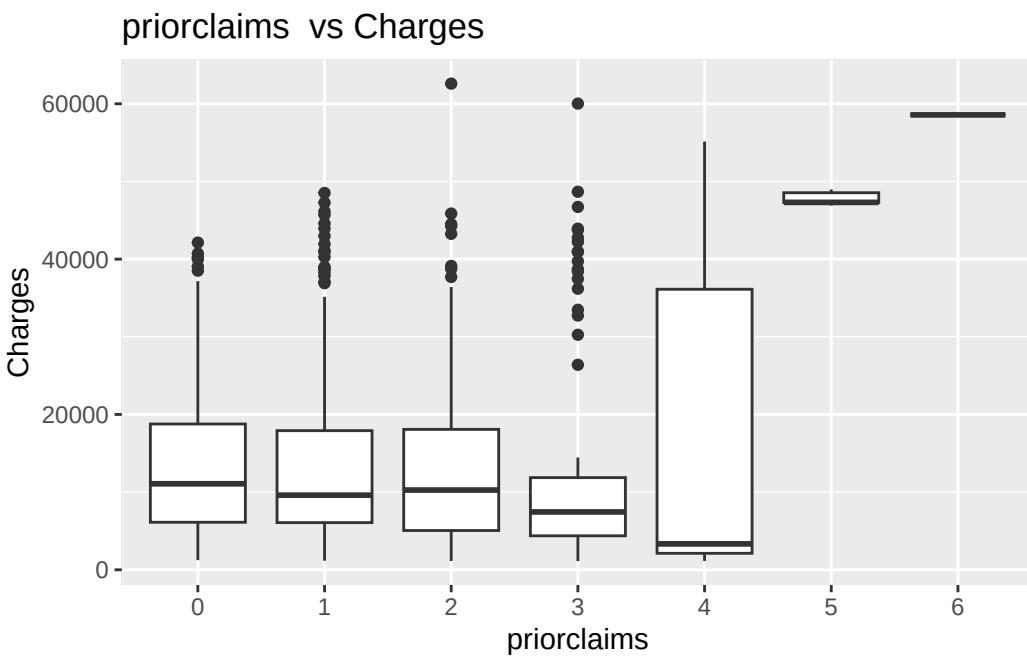

```
Warning: Use of `insurance_data[[col_name]]` is discouraged.
i Use `.data[[col_name]]` instead.
```




```
Warning: Use of `insurance_data[[col_name]]` is discouraged.
i Use `.data[[col_name]]` instead.
```



```
Warning: Use of `insurance_data[[col_name]]` is discouraged.
i Use `.data[[col_name]]` instead.
```



Modeling Methods

```r
# Train/test split using seed 288
set.seed(288)

train_indices <- sample(nrow(insurance_data), floor(nrow(insurance_data) * 0.85))

train <- insurance_data[train_indices, ]
test <- insurance_data[-train_indices, ]
```

```r
# Full model with all predictors
full_model <- lm(charges ~ ., data = train)

# Stepwise selection using AIC
step_model <- step(full_model, direction = "backward", k = 2, trace = FALSE)

# Show summaries for both models
summary(full_model)
```

```
Call:
lm(formula = charges ~ ., data = train)

Residuals:
   Min     1Q Median     3Q    Max 
-10345  -3216  -1342   1254  25169 

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)    
(Intercept)     -1.105e+04  1.564e+03  -7.064 3.82e-12 ***
X               -4.721e-01  9.226e-01  -0.512  0.60903    
caseID          -2.565e-01  5.984e-01  -0.429  0.66827    
age              2.530e+02  1.658e+01  15.254  < 2e-16 ***
sexmale         -2.939e+02  4.602e+02  -0.639  0.52326    
bmi              3.527e+02  3.993e+01   8.832  < 2e-16 ***
children         4.578e+02  1.917e+02   2.388  0.01719 *  
smokeryes        2.211e+04  5.638e+02  39.224  < 2e-16 ***
regionnorthwest -8.324e+02  6.587e+02  -1.264  0.20672    
regionsoutheast -2.212e+03  6.717e+02  -3.293  0.00104 ** 
regionsouthwest -1.600e+03  6.629e+02  -2.414  0.01602 *  
month           -1.154e+02  6.664e+01  -1.732  0.08371 .  
priorclaims      4.931e+02  1.906e+02   2.587  0.00988 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6169 on 718 degrees of freedom
Multiple R-squared:  0.7215,	Adjusted R-squared:  0.7168 
F-statistic:   155 on 12 and 718 DF,  p-value: < 2.2e-16
```

```r
summary(step_model)
```

```
Call:
lm(formula = charges ~ age + bmi + children + smoker + region + 
    month + priorclaims, data = train)

Residuals:
   Min     1Q Median     3Q    Max 
 -9934  -3239  -1342   1326  25398 

Coefficients:
```

```
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)     -11545.59    1426.79  -8.092 2.49e-15 ***
age                253.70      16.53  15.346  < 2e-16 ***
bmi                351.27      39.72   8.844  < 2e-16 ***
children           456.19     191.37   2.384   0.0174 *
smokeryes        22077.38     561.40  39.326  < 2e-16 ***
regionnorthwest   -837.06     657.19  -1.274   0.2032
regionsoutheast  -2213.17     670.08  -3.303   0.0010 **
regionsouthwest  -1582.54     660.97  -2.394   0.0169 *
month             -114.25      66.37  -1.721   0.0856 .
priorclaims        483.79     189.78   2.549   0.0110 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6160 on 721 degrees of freedom
Multiple R-squared:  0.7211,    Adjusted R-squared:  0.7177
F-statistic: 207.2 on 9 and 721 DF,  p-value: < 2.2e-16
```

```r
# Compare model fit statistics
AIC(full_model)
```

```
[1] 14848.69
```

```r
AIC(step_model)
```

```
[1] 14843.56
```

```r
BIC(full_model)
```

```
[1] 14913.01
```

```r
BIC(step_model)
```

```
[1] 14894.1
```

Refinements

```r
# Backwards BIC on the AIC model
bic_step_model <- step(step_model, direction = "backward", k = log(nrow(train)),
                       trace = FALSE)

# Show summaries for both models
summary(step_model)
```

```
Call:
lm(formula = charges ~ age + bmi + children + smoker + region +
    month + priorclaims, data = train)

Residuals:
   Min     1Q Median     3Q    Max
 -9934  -3239  -1342   1326  25398

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)     -11545.59    1426.79  -8.092 2.49e-15 ***
age                253.70      16.53  15.346  < 2e-16 ***
bmi                351.27      39.72   8.844  < 2e-16 ***
children           456.19     191.37   2.384   0.0174 *
smokeryes        22077.38     561.40  39.326  < 2e-16 ***
regionnorthwest   -837.06     657.19  -1.274   0.2032
regionsoutheast  -2213.17     670.08  -3.303   0.0010 **
regionsouthwest  -1582.54     660.97  -2.394   0.0169 *
month             -114.25      66.37  -1.721   0.0856 .
priorclaims        483.79     189.78   2.549   0.0110 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6160 on 721 degrees of freedom
Multiple R-squared:  0.7211,    Adjusted R-squared:  0.7177
F-statistic: 207.2 on 9 and 721 DF,  p-value: < 2.2e-16
```

```r
summary(bic_step_model)
```

```
Call:
lm(formula = charges ~ age + bmi + smoker, data = train)
```

```
Residuals:
   Min     1Q Median     3Q    Max 
-10172  -3349  -1130   1152  27588 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -11357.41    1304.61  -8.706   <2e-16 ***
age            253.15      16.62  15.228   <2e-16 ***
bmi            323.08      38.22   8.453   <2e-16 ***
smokeryes    22185.63     566.89  39.136   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6248 on 727 degrees of freedom
Multiple R-squared:  0.7107,	Adjusted R-squared:  0.7095 
F-statistic: 595.4 on 3 and 727 DF,  p-value: < 2.2e-16
```

```r
# Compare model fit statistics
AIC(step_model)
```

```
[1] 14843.56
```

```r
AIC(bic_step_model)
```

```
[1] 14858.34
```

```r
BIC(step_model)
```

```
[1] 14894.1
```

```r
BIC(bic_step_model)
```

```
[1] 14881.31
```

```r
summary(step_model)$r.squared
```

```
[1] 0.721134
```

```r
summary(bic_step_model)$r.squared
```

```
[1] 0.7107277
```

```r
# Test prep
y_train <- train$charges
x_train <- select(train, -charges)
y_test <- test$charges
x_test <- select(test, -charges)

RMSE <- function(model) {
  x_test <- select(test, -charges)
  predicted <- predict(model, newdata = x_test)
  sqrt(mean(predicted - y_test)^2)
}

# ...and Final Results

aic_rmse <- RMSE(step_model)

RMSE(full_model)
```

```
[1] 1387.941
```

```r
aic_rmse
```

```
[1] 1395.945
```

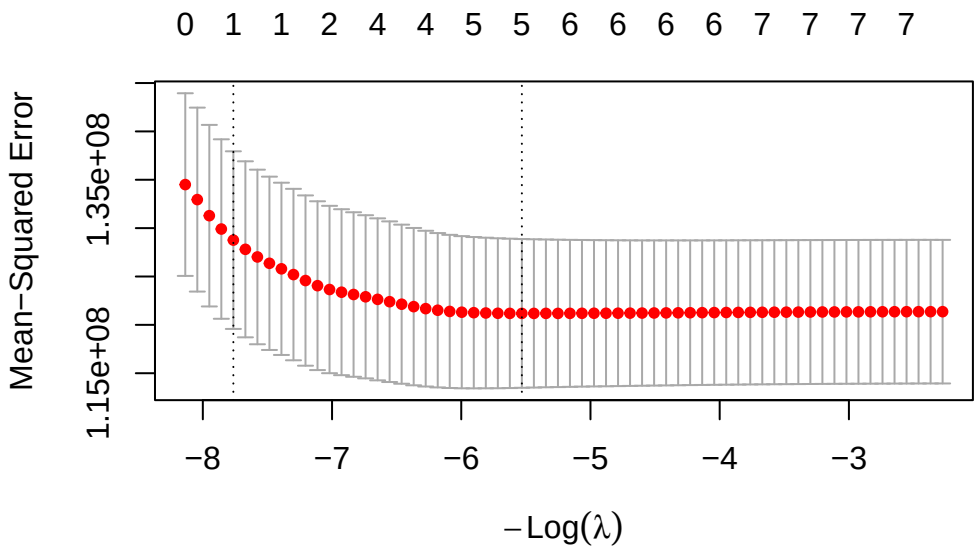
```r
RMSE(bic_step_model)
```

```
[1] 1606.86
```

```r
# LASSO Prep

x_train <- as.matrix(x_train)
x_test <- as.matrix(x_test)

# LASSO Regression

cv_model <- cv.glmnet (x = x_train, y = y_train, alpha = 1, nfolds = 10,
                       type.measure = "mse")
```

```
Warning in storage.mode(x) <- "double": NAs introduced by coercion
Warning in storage.mode(x) <- "double": NAs introduced by coercion
Warning in storage.mode(x) <- "double": NAs introduced by coercion
Warning in storage.mode(x) <- "double": NAs introduced by coercion
Warning in storage.mode(x) <- "double": NAs introduced by coercion
Warning in storage.mode(x) <- "double": NAs introduced by coercion
Warning in storage.mode(x) <- "double": NAs introduced by coercion
Warning in storage.mode(x) <- "double": NAs introduced by coercion
Warning in storage.mode(x) <- "double": NAs introduced by coercion
Warning in storage.mode(x) <- "double": NAs introduced by coercion
Warning in storage.mode(x) <- "double": NAs introduced by coercion

Warning in cbind2(1, newx) %*% nbeta: NAs introduced by coercion
Warning in cbind2(1, newx) %*% nbeta: NAs introduced by coercion
Warning in cbind2(1, newx) %*% nbeta: NAs introduced by coercion
Warning in cbind2(1, newx) %*% nbeta: NAs introduced by coercion
Warning in cbind2(1, newx) %*% nbeta: NAs introduced by coercion
Warning in cbind2(1, newx) %*% nbeta: NAs introduced by coercion
Warning in cbind2(1, newx) %*% nbeta: NAs introduced by coercion
Warning in cbind2(1, newx) %*% nbeta: NAs introduced by coercion
Warning in cbind2(1, newx) %*% nbeta: NAs introduced by coercion
Warning in cbind2(1, newx) %*% nbeta: NAs introduced by coercion
```

```r
# Outputs

plot(cv_model)
```




```r
print(cv_model$lambda.min)
```

```
[1] 252.3761
```

```r
print(cv_model$lambda.1se)
```

```
[1] 2353.668
```

```r
print(coef(cv_model, s = "lambda.min"))
```

```
11 x 1 sparse Matrix of class "dgCMatrix"
             lambda.min
(Intercept) -2275.55555
X                  .
caseID             .
age            219.50526
sex                .
bmi            180.06158
children       605.97080
smoker             .
```

```
region          .
month         -55.66982
priorclaims   629.61343
```

```r
print(coef(cv_model, s = "lambda.1se"))
```

```
11 x 1 sparse Matrix of class "dgCMatrix"
             lambda.1se
(Intercept) 10160.86471
X               .
caseID          .
age            75.77773
sex             .
bmi             .
children        .
smoker          .
region          .
month           .
priorclaims     .
```

```r
print("RMSE values for the three candidate models")
```

```
[1] "RMSE values for the three candidate models"
```

```r
table1 <- data.frame("Model" = c("Backward AIC", "1-SE", "MSE"), "RMSE" =
                     c(aic_rmse, sqrt(mean((predict(cv_model, s = "lambda.1se",
                     newx = x_test) - y_test)^2)), sqrt(mean((predict(cv_model,
                     s = "lambda.min", newx = x_test) - y_test)^2))))
```

```
Warning in cbind2(1, newx) %*% nbeta: NAs introduced by coercion
Warning in cbind2(1, newx) %*% nbeta: NAs introduced by coercion
```

```r
table1
```

```
         Model      RMSE
1 Backward AIC  1395.945
2         1-SE 13072.044
3          MSE 12653.628
```

```r
null_model <- lm(charges ~ 1, data = train)

# Forward Models

forward_bic_model <- step(null_model, direction = "forward", k = log(nrow(train)),
                     scope = list(lower = ~1, upper = full_model), trace = FALSE)
forward_aic_model <- step(null_model, direction = "forward", k = 2, scope =
                            list(lower = ~1, upper = full_model), trace = FALSE)

print("Compare forward BIC and AIC")
```

```
[1] "Compare forward BIC and AIC"
```

```r
# Outputs - AIC performs best

summary(step_model)
```

```
Call:
lm(formula = charges ~ age + bmi + children + smoker + region +
    month + priorclaims, data = train)

Residuals:
   Min     1Q Median     3Q    Max
 -9934  -3239  -1342   1326  25398

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)     -11545.59    1426.79  -8.092 2.49e-15 ***
age                253.70      16.53  15.346  < 2e-16 ***
bmi                351.27      39.72   8.844  < 2e-16 ***
children           456.19     191.37   2.384   0.0174 *
smokeryes        22077.38     561.40  39.326  < 2e-16 ***
regionnorthwest   -837.06     657.19  -1.274   0.2032
regionsoutheast  -2213.17     670.08  -3.303   0.0010 **
regionsouthwest  -1582.54     660.97  -2.394   0.0169 *
month             -114.25      66.37  -1.721   0.0856 .
priorclaims        483.79     189.78   2.549   0.0110 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 6160 on 721 degrees of freedom
Multiple R-squared:  0.7211,    Adjusted R-squared:  0.7177
F-statistic: 207.2 on 9 and 721 DF,  p-value: < 2.2e-16
```

```r
summary(forward_bic_model)
```

```
Call:
lm(formula = charges ~ smoker + age + bmi, data = train)

Residuals:
   Min     1Q Median     3Q    Max 
-10172  -3349  -1130   1152  27588 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -11357.41    1304.61  -8.706   <2e-16 ***
smokeryes    22185.63     566.89  39.136   <2e-16 ***
age            253.15      16.62  15.228   <2e-16 ***
bmi            323.08      38.22   8.453   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6248 on 727 degrees of freedom
Multiple R-squared:  0.7107,    Adjusted R-squared:  0.7095 
F-statistic: 595.4 on 3 and 727 DF,  p-value: < 2.2e-16
```

```r
summary(forward_aic_model)
```

```
Call:
lm(formula = charges ~ smoker + age + bmi + region + priorclaims + 
    children + month, data = train)

Residuals:
   Min     1Q Median     3Q    Max 
 -9934  -3239  -1342   1326  25398 

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)    
(Intercept)     -11545.59    1426.79  -8.092 2.49e-15 ***
```

```
smokeryes        22077.38     561.40  39.326  < 2e-16 ***
age                253.70      16.53  15.346  < 2e-16 ***
bmi                351.27      39.72   8.844  < 2e-16 ***
regionnorthwest   -837.06     657.19  -1.274   0.2032
regionsoutheast  -2213.17     670.08  -3.303   0.0010 **
regionsouthwest  -1582.54     660.97  -2.394   0.0169 *
priorclaims        483.79     189.78   2.549   0.0110 *
children           456.19     191.37   2.384   0.0174 *
month             -114.25      66.37  -1.721   0.0856 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6160 on 721 degrees of freedom
Multiple R-squared:  0.7211,    Adjusted R-squared:  0.7177
F-statistic: 207.2 on 9 and 721 DF,  p-value: < 2.2e-16
```

```r
paste("Forward BIC RMSE", RMSE(forward_bic_model))
```

```
[1] "Forward BIC RMSE 1606.85983569018"
```

```r
paste("Forward AIC RMSE", RMSE(forward_aic_model))
```

```
[1] "Forward AIC RMSE 1395.94473046712"
```

```r
print("Compare forward AIC to backward AIC")
```

```
[1] "Compare forward AIC to backward AIC"
```

```r
print("AIC -----")
```

```
[1] "AIC -----"
```

```r
paste("Backward AIC", AIC(step_model))
```

```
[1] "Backward AIC 14843.5607905206"
```

```r
paste("Forward AIC", AIC(forward_aic_model))
```

```
[1] "Forward AIC 14843.5607905206"
```

```r
print("BIC -----")
```

```
[1] "BIC -----"
```

```r
paste("Backward AIC", BIC(step_model))
```

```
[1] "Backward AIC 14894.0993385778"
```

```r
paste("Forward AIC", BIC(forward_aic_model))
```

```
[1] "Forward AIC 14894.0993385778"
```

```r
print("R^2 -----")
```

```
[1] "R^2 -----"
```

```r
paste("Backward AIC", summary(step_model)$r.squared)
```

```
[1] "Backward AIC 0.72113397002663"
```

```r
paste("Forward AIC", summary(forward_aic_model)$r.squared)
```

```
[1] "Forward AIC 0.72113397002663"
```

```r
print("RMSE -----")
```

```
[1] "RMSE -----"
```

```r
paste("Backward AIC", aic_rmse)
```

```
[1] "Backward AIC 1395.94473046713"
```

```r
paste("Forward AIC", RMSE(forward_aic_model))
```

```
[1] "Forward AIC 1395.94473046712"
```

RMSE On New Data

```r
insurance_data_validation <- drop_na(read.csv("insurance_data_validation_S26", header = TRUE)

RMSE <- function(model) {
  x_test <- select(insurance_data_validation, -charges)
  y_test <- insurance_data_validation$charges
  predicted <- predict(model, newdata = x_test)
  sqrt(mean(predicted - y_test)^2)
}

RMSE(step_model)
```

```
[1] 112.6422
```